# SAVER-VLA: Feedback-Driven Agentic Failure Recovery for Vision-Language-Action Manipulation

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**Abstract:** Vision-language-action (VLA) policies handle nominal manipulation well, but after a slip or displacement they keep executing actions that no longer apply. SAVER-VLA is an agentic layer that lets a pretrained VLA (SmolVLA) detect, diagnose, and recover from such failures without retraining. A verifier flags the failure, an LLM agent selects bounded recovery skills one at a time, and each measured outcome is fed back so the agent can replan or return control to the VLA. In LIBERO fault-injection experiments, the agent restores task success under object displacement, gripper release, actuator impairment, and approach miscalibration, adapting its strategy to the evidence without knowing the fault.

**Keywords:** Agentic robotics, VLA models, Failure recovery, LLM agents

**Project page:** https://mehdiomid.github.io/saver-vla/

## 1 Introduction

VLA models map images and instructions directly to robot actions, yet they have no explicit notion of failure. A recent benchmark shows that state-of-the-art VLAs that succeed on nearly all nominal LIBERO tasks lose more than half their performance once a real execution failure occurs [1]. Language models can plan over skills, explain or correct failures [2–3], and decompose tasks for a VLA [4]. However, this reasoning is rarely used to supervise a pretrained VLA online, during its own execution.

We ask whether an LLM agent, invoked only after a detected failure, can guide a pretrained VLA through recovery: choosing bounded skills, interpreting their outcomes, and deciding when to hand control back. SAVER-VLA does this without retraining. We contribute (i) an event-triggered hierarchy for agent-supervised recovery, (ii) feedback that separates *target reached* from *grasp verified*, and (iii) a fault-injection study scoring decisions, skills, and task success separately.

## 2 Method: Detect, Diagnose, Act, Verify, Replan

**Two loops at two time scales** (Fig. 1a). SmolVLA (via LeRobot) drives a simulated Panda arm at 20 Hz. A verifier checks grasp and transport predicates every step; only on a failure is the agent called, and again after each skill or policy trial. The agent never outputs motor commands, keeping fast control and slow reasoning decoupled.

**Agent interface.** Each decision receives a JSON observation (grasp state, end-effector object offset, last skill and trial outcomes, gripper-command mismatch, and available and blocked skills), never the fault's name. The agent (GPT-5.2 or GPT-5.5) returns a skill, variant, and reason, so every action has a readable rationale.

**Skills.** Motion skills (pregrasp, approach, ±X grasp offset, 2D repositioning) issue bounded commands with timeouts and report `REACHED` or `FAILED`; reassess and abort complete the library. `RESUME_POLICY` starts a bounded SmolVLA trial that the verifier labels `VERIFIED` or `NOT_VERIFIED`: the agent positions the robot, and the learned policy performs the grasp.

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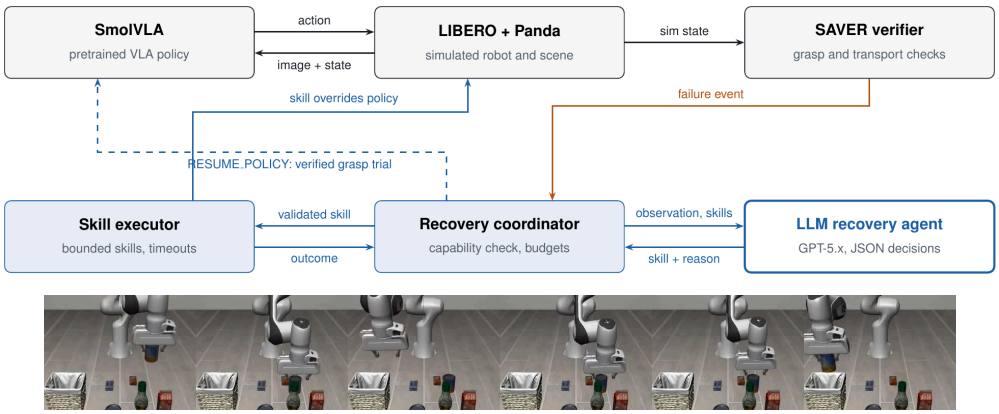



**Figure 1: (a, top)** Control loop and event-triggered recovery loop with outcome feedback. **(b, bottom)** Recovery trace: after a centered grasp fails with no gripper mismatch, GPT-5.5 agent applies a −X offset (5.9 → 1.3 cm) and the next trial is verified.

## 3 Experiments and Results

**Setup.** We use `lerobot/smolvla_libero` on LIBERO-Object task 0 (soup can to basket) in MuJoCo with privileged state. At step 67 we displace the can by 12 cm (*separation*) or force the gripper open (*release*); harder variants impair the gripper during the first resumed trial or add a 3.5 or 5.5 cm +X approach bias.

| Run | Fault | Controller/Agent | Agent decisions | Agent | Skills | Task |
|---|---|---|---|---|---|---|
| Baseline | Separation | SmolVLA only | – | – | – | ✗ |
| Baseline | Separation | Queue reset | – | – | – | ✗ |
| Scripted | Separation | Fixed sequence | – | – | ✓ | ✓ |
| P1 | Separation | GPT-5.5 | 3 | ✓ | ✓ | ✓ |
| P2 | Release | GPT-5.2 | 9 (budget hit) | ✗ | ✓ | ✓† |
| P2 | Release | GPT-5.5 | 3 | ✓ | ✓ | ✓ |
| P3 | Release + impairment | GPT-5.5 | 7 | ✓ | - | ✓ |
| P4 | Release + bias | GPT-5.5 | request overridden | ✓ | ✗ | ✓† |
| P5 | Release + bias | GPT-5.5 | 8 | ✓ | ✗ | ✗ |
| P6 | Release + bias | GPT-5.5 | 8 | ✓ | - | ✓ |

**Table 1:** Agent decisions and success rate

**SAVER-VLA restores task success under every fault family tested** (Table 1). SmolVLA alone and a policy-queue reset never return to the dropped can, while the agent recovers under all four fault types. On the simplest fault it finds the sequence an expert would script; on harder faults it goes beyond it.

**Reasoning is lightweight and targeted.** GPT-5.5 agent is consulted only 3–8 times per episode (about 1% of control steps) and recovers from gripper release with three decisions.

**Reasoning quality determines the plan.** At the same near-grasp state, GPT-5.5 hands control to SmolVLA (3 decisions) and successfully completes the task while GPT-5.2 reads the not-yet-attempted grasp as a failed approach and repeats repositioning until its budget runs out. GPT-5.2 agent mistook a not-yet-attempted grasp for a failed one.

**Feedback drives evidence-grounded replanning.** When a grasp trial ends `NOT_VERIFIED`, the agent infers the cause from feedback alone. If the gripper did not follow its commands (P3), it attributes the failure to actuation and retries the same centered grasp. If the gripper worked but the hand sat 5.9 cm off target (P6, Fig. 1b), it shifts the grasp in −X instead. A fixed script cannot make this distinction, because it repeats the same grasp regardless of the cause.

**Capability-aware agency grows with the skill library.** Shown blocked skills, the agent aborts on its own when no action is feasible. When 2D repositioning stopped 21.4 cm from the can (P5), a height-aware version (P6) turned the same reasoning into success: new skills extend the agent without changing it.

## References


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